OAI CN & RAN Overview

Open6G+AI school

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March 30, 2026

Outline

CN Overview & Features

gNB Overview & Features

gNB Deployment examples

UE Overview & Features

gNB/UE Radios and Accelerators

gNB/UE Support for NTN

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General Feature Set

Basic procedures

- Connection and registration procedures
  - Compliant with 3GPP Rel. 16, evolving toward Rel. 17+
  - Multiple UEs
  - Multiple PDU sessions per UE
  - Service Request
- PDU Session management
  - Establishment
  - Modification
  - Release

Additional Features

- End-to-end QoS for control and user planes
- Ethernet PDU Sessions for Time-Sensitive Networking (TSN)
- Framed routing, location services, and network analytics (NWDAF)
- Event exposure (NEF), N2 handover, network slicing, multiple UPFs, and ULCL support
- N2 handover procedures
- Network slicing (NSSF, multiple AMFs)
- Multiple UPFs

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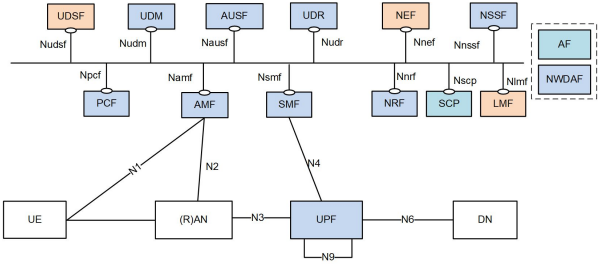
Deployment flavors & options

Modes of OAI 5G core

- Minimalist 5GC: AMF, SMF, UPF, and NRF
- Basic 5GC: adds UDM, AUSF, UDR
- Full 5GC: PCF, NSSF, NEF (incomplete)
- UPF in two modes:
  - Simple switch (based on SPGW-U/4G)
  - eBPF/XDP-based



Deployment Options

- Automated deployment of NFs in Docker containers through docker compose
- Cloud-native using Helm Charts in OpenShift Cluster

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Interworking and Testing

- Interworking with gNBs:
  - OAI gNB
  - Amarisoft
  - Baicells
- Interworking with Rel. 15&16 (COTS) UEs
  - Quectel/SIMcom modules
  - iPhone 14-17
  - Huawei P40/P40 Pro
  - Pixel 5+
  - One Plus 8
  - Amarisoft UE
  - OAI UE
- Validation of all NFs with CI/CD pipeline
  - Functional, stability, reliability and performance tests
  - Automated integration tests and unit tests (on roadmap)
- Open-source RAN simulators
  - gnbsim
  - UERANSIM
  - My5g-RANTester
  - Omec-gnbsim

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gNB General Feature Set

OAI gNB (CU, DU) – interoperable with COTS equipment

Radio

- FR1, FR2 (experimental)
- Bandwidths: 10-200MHz; 256-QAM
- MIMO: DL 4 layers, UL 2 layers
- Throughput: 1.4 Gbps (DL), 400 Mbps (UL), depending on TDD pattern
- Latency: down to 5 ms RTT (application, depends on TDD config)

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gNB General Feature Set

Functional

- Up to 64 UEs* supported
- Support for 3GPP/O-RAN/SCF interfaces
  - Multiple DUs and CU-UPs at one CU-CP
  - E2 agent through FlexRIC
  - Front-/Midhaul
  - O1 interface through O1 adapter
- Inline accelerator: Nvidia Aerial L1 Offload
- Look-aside accelerators (see later)
- NTN LEO/GEO, RedCap initial support

Tested (COTS) UEs (selection)

- Quectel modems
- Samsung S25
- iPhone 14, 15
- Pixel 8, 9, 10

Tested 5GC

- OAI 5GC
- Open5GS, Free5GC
- SABOX, Amarisoft, others…

Complete Feature Set: doc/FEATURE_SET.md (in the repository)

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gNB Performance

gNB – **Max Downlink Throughput**, Band n41, TDD: DDDFU

DL		Bandwidth (MHz)/Number of PRBs @ 30kHz											
(Mbps)		10/24	15/38	20/51	25/65	30/78	40/106	50/133	60/162	70/189	80/217	90/245	100/273
Layers	1			**72**			**152**		**233**		**310**		**400**
	2			**143**			**304**		**466**		**622**		**800**
	4			**258**			**550**		**730**				**1400***

gNB – **Max Uplink Throughput**, Band n41, TDD: DDFUU

UL		Bandwidth (MHz)/Number of PRBs @ 30kHz											
(Mbps)		10/24	15/38	20/51	25/65	30/78	40/106	50/133	60/162	70/189	80/217	90/245	100/273
Layers	1			**39**			**81**		**123**		**80**		**101**
	2			**65**			**154**		**175**		**140**		**130**
	4												

(Measurements in Allbesmart lab, gNB/N310+`--tune-offset`, UE/Quectel RM500Q-GL; 100MHz 4-layer w/ VVDN O-RU)

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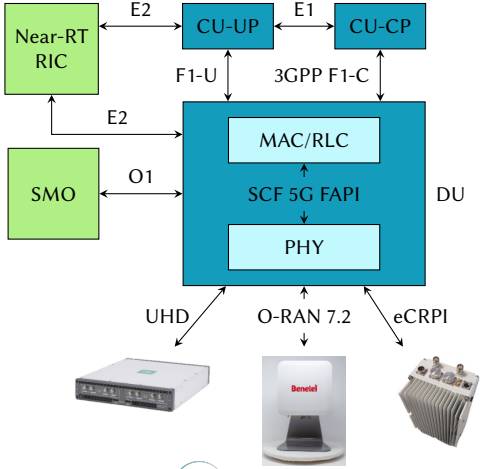
OpenAirInterface and O-RAN/3GPP/SCF



- **O1**: Integration with ONAP-based SMO
- **E2**: Supported through E2 Agent and FlexRIC
- **E1**: Multiple CU-UPs per CU-CP
- **F1**: Multiple DUs per CU
- **FAPI**: all* L1 procedures SCF 5G FAPI 222.10.02; Nvidia CUDA-Accel. RAN interoperability; nFAPI
- **Fronthaul**: various split 7.2/split 8 options, see later

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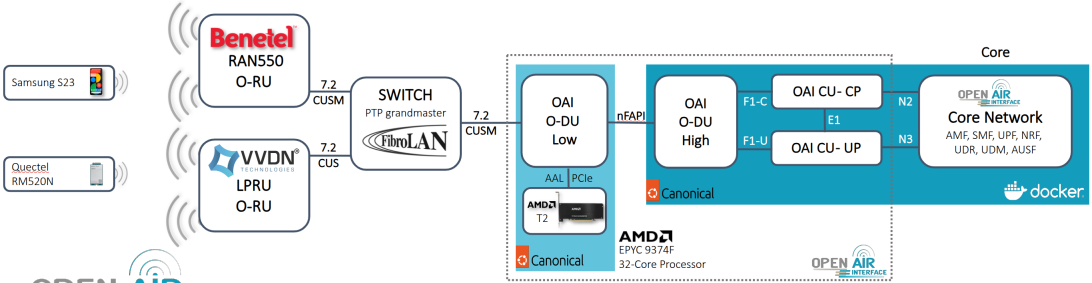
O-RAN 7.2 Fronthaul Setup with T2-Telco Card

- 3GPP F1 and E1 midhaul splits between OAI CU-CP/CU-UP/DU
- SCF nFAPI interface between O-DU Low/O-DU High
- Optionally: AMD T2 Lookaside Accelerator card
- O-RAN 7.2 fronthaul split using OSC fronthaul interface library (FHI, F release)
- 1.4 Gbps throughput in DL



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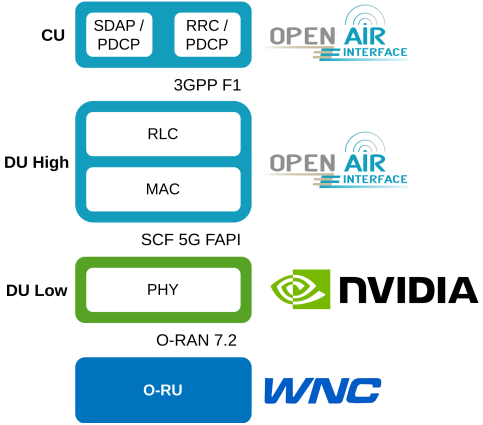
Nvidia CUDA-Accelerated RAN Setup



- Aerial CUDA-Accelerated RAN: GPU is inline L1 accelerator
- L2/L3 from OpenAirInterface, Interface via SCF 5G FAPI
- DU Layer 1 southbound interface: O-RAN 7.2, WNC O-RU
- DU Layer 2 northbound interface: 3GPP F1, E1
- 1 Gbps+ Downlink (4-layer MIMO) and 300 Mbps+ Uplink

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UE General Feature Set

OAI UE – (partially) interoperable with COTS equipment

Radio

- FR1, FR2 (experimental)
- Bandwidths: 10-100MHz; 256-QAM
- Support for SISO, MIMO 2 layers UL/DL experimental
- Throughput: up to 230 Mbps (DL), up to 140 Mbps (UL)
- Latency: limited minimum RX-to-TX latency (3)

Complete Feature Set: doc/FEATURE_SET.md (in the repository)

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OAI UE General Feature set

Functional

- Can support up to 10 UEs per process (Only RFsim)
- Uses FAPI-inspired L1/L2 interface ("UE FAPI")
- RRC Connection Setup/Reestablishment/Measurements/HO
- IP/Ethernet PDU sessions
- Look-aside accelerators (see later)
- NTN LEO/GEO

RAN Interoperability

- OAI gNB
- Nvidia CUDA-Accel. RAN (L1)
- srsRAN_project/OCUDU
- COTS: Ericsson, Nokia (partially)

Complete Feature Set: doc/FEATURE_SET.md (in the repository)

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OAI UE Performance (2026)

- Measurements on Nvidia DGX Spark (ARM), but translate to x86 as well
- Using N310/x400 with UHD 4.9 and DPDK 24.11
- 100 MHz: max DL 230 Mbps, UL 118 Mbps (SISO)
- 40 MHz: max DL 85 Mbps, UL 46 Mbps (SISO)

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Radio Platforms

- O-RAN 7.2 CUS-plane (with Benetel, LITEON, VVDN, Metanoia, Foxconn), *CUSM*-plane with Benetel [gNB only]
- eCPRI with AW2S (split 8) [gNB only]
- USRP through UHD (split 8): B210, N3xx, x400
- BladeRF (split 8)
- LimeSDR (split 8)
- Simulation: RFsim (non-RT), vrtsim (RT), ZeroMQ (non-RT)

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Accelerators

In-line

- Nvidia CUDA-Accelerated RAN (see also separate slide earlier), Documentation

Look-aside: O-RAN AAL

- O-RAN AAL (bbdev-based): AMD/Xilinx T2 card, Intel ACC200/vRAN Boost; Documentation
- Xilinx PCIe-XDMA based FPGA LDPC, Documentation
- CUDA-based GPU LDPC Offload: under integration, see here
- Accelerators can be used in simulators or softmodems (UE/gNB, any radio)

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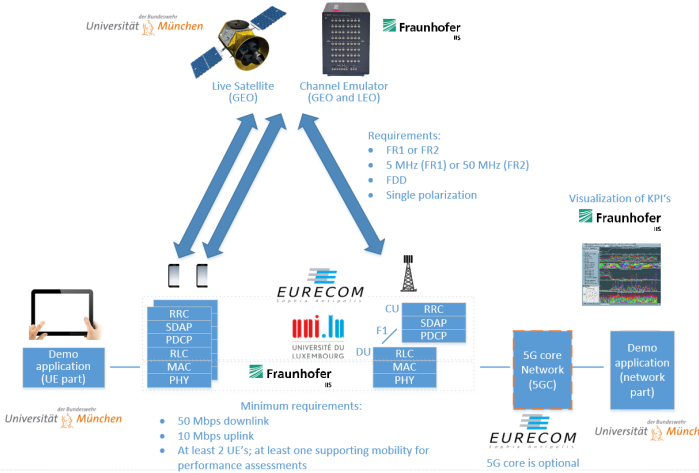
ESA 5G-GOA/5G-LEO

- Forward data over relay GEO satellite, or emulator (LEO)
- Continuous freq. offset compensation, UL Timing
- SNR measurement, CSI Reporting
- AMC without HARQ feedback
- Handover



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NTN Feature Set

See also OAI Feature Set with more details

Supported at gNB/UE

- Support for transmission/reception of SIB19
- HARQ improvements: no HARQ (for GEO) and 32 HARQ (for LEO)
- SS-SINR measurement and CSI reporting
- RFsim simulation of GEO and LEO channels
- CI RFsim tests for GEO and LEO scenarios
- Test with channel emulators by external parties

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NTN Feature Set (Continued)

Further features at UE

- Continuous Frequency Offset & Compensation for LEO scenarios
- Timing adjustments based on ntn-Config information
- DL/UL doppler (pre-)compensation

Supported NTN bands

- FR1: 255, 256 (Rel 17), 254 (Rel 18)
- FR2: 510, 511, 512 (Rel 18)

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OAI: What & Where

- Get the RAN at `https://gitlab.eurecom.fr/oai/openairinterface5g/`
- Includes tutorials and documentation
- All development of OAI RAN happens here!
- We invite partners to our Slack channel: `https://openairinterface.slack.com/`
- There are public mailing lists to ask user questions and get help from the community!
- A number of companies provide ready-to-go setups and support.

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Thank you!

Q? A!

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